

第十三章：《古典概型与古典概型概率计算》习题

Exercises on Classical Probability Model and Its Probability Calculation

一、选择题（共 20 题）

I. Multiple-Choice Questions (20 questions in total)

1. 某人打靶时连续射击两次，下列事件中与事件“至少一次中靶”互为对立的是（ ）

A person shoots twice in a row at a target. Which of the following events is complementary to the event "at least one hit"? ()

A. 至多一次中靶 B. 两次都中靶 C. 只有一次中靶 D. 两次都没有中靶

A. At most one hit B. Two hits C. Exactly one hit D. No hits

2. 从个位数与十位数之和为奇数的两位数中任取一个，其个位数为 0 的概率是（ ）

Take a random two-digit number where the sum of the units digit and the tens digit is odd. What is the probability that its units digit is 0? ()

A. $\frac{4}{9}$ B. $\frac{1}{3}$ C. $\frac{2}{9}$ D. $\frac{1}{9}$

3. 同时抛掷两枚质地均匀的硬币，出现两枚都是正面朝上的概率为（ ）

Toss two fair coins simultaneously. What is the probability that both land heads up? ()

A. $\frac{1}{4}$ B. $\frac{1}{3}$ C. $\frac{1}{2}$ D. $\frac{3}{4}$

4. 国庆阅兵中，某兵种 A, B, C 三个方阵按随机次序通过主席台，则 B 先于 A, C 通过的概率为（ ）

In the National Day military parade, three phalanxes A, B, and C of a certain arm pass the rostrum in random order. What is the probability that B passes before A and C? ()

A. $\frac{1}{6}$ B. $\frac{1}{3}$ C. $\frac{1}{2}$ D. $\frac{2}{3}$

5. 甲、乙两位同学考入同一专业（设 3 个班级），他们被随机分到同一个班级的概率为（ ）

Two students, A and B, are admitted to the same major (with 3 classes). What is the probability that they are randomly assigned to the same class? ()

A. $\frac{1}{9}$ B. $\frac{1}{6}$ C. $\frac{1}{3}$ D. $\frac{1}{2}$

6. 从 20 名男同学、10 名女同学中任选 3 名参加体能测试，选到的 3 名同学中既有男同学又有女同学的概率为（ ）

Select 3 students randomly from 20 male students and 10 female students to participate in a physical fitness test. What is the probability that the selected 3 students include both male and female students? ()

A. $\frac{20}{29}$ B. $\frac{19}{29}$ C. $\frac{10}{29}$ D. $\frac{9}{29}$

7. 有 5 本不同的书（2 本语文、2 本数学、1 本物理），随机并排摆放在书架上，同一科目的书都不相邻的概率为（ ）

There are 5 different books (2 Chinese, 2 Math, and 1 Physics). They are randomly placed side

by side on a bookshelf. What is the probability that books of the same subject are not adjacent?
()

A. $\frac{1}{5}$ B. $\frac{2}{5}$ C. $\frac{3}{5}$ D. $\frac{4}{5}$

8. 一个盒子有 6 个小格子, 现有桔子、苹果、香蕉各 2 个, 将其随机放入格子 (每个格子 1 个), 放好后每行、每列水果种类各不相同的概率是 ()

A box has 6 small grids. There are 2 oranges, 2 apples, and 2 bananas. They are randomly placed into the grids (1 per grid). What is the probability that the types of fruits in each row and column are different after placement? ()

A. $\frac{2}{15}$ B. $\frac{2}{9}$ C. $\frac{1}{5}$ D. $\frac{1}{3}$

9. 铸币大臣在每箱 100 枚硬币中掺入 1 枚劣币, 国王用两种方法检测: 方法一 (10 箱各抽 1 枚)、方法二 (5 箱各抽 2 枚), 能发现至少 1 枚劣币的概率分别为 p_1, p_2 , 则 ()

The Minister of Coinage mixes 1 counterfeit coin into each box of 100 coins. The king uses two detection methods: Method 1 (draw 1 coin from each of 10 boxes) and Method 2 (draw 2 coins from each of 5 boxes). The probabilities of finding at least 1 counterfeit coin are p_1 and p_2 respectively. Then ()

A. $p_1 = p_2$ B. $p_1 < p_2$ C. $p_1 > p_2$ D. 无法确定

A. $p_1 = p_2$ B. $p_1 < p_2$ C. $p_1 > p_2$ D. Cannot be determined

10. 从三男三女 6 名学生中任选 2 名 (每名被选中机会相等), 则 2 名都是女同学的概率等于 ()

Select 2 students randomly from 6 students (3 boys and 3 girls) with equal chance of being selected. What is the probability that both selected students are female? ()

A. $\frac{1}{5}$ B. $\frac{1}{3}$ C. $\frac{1}{2}$ D. $\frac{2}{3}$

11. 从 n 个正整数 $1, 2, \dots, n$ 中任意取出两个不同的数, 若两数之和等于 5 的概率为 $\frac{1}{14}$, 则 $n =$ ()

Take two different numbers randomly from n positive integers $1, 2, \dots, n$. If the probability that the sum of the two numbers is 5 is $\frac{1}{14}$, then $n =$ ()

A. 6 B. 7 C. 8 D. 9

12. 将 3 个相同黑球和 3 个相同白球自左向右排成一排, 若“从任意位置向左数, 黑球个数 $\geq$ 白球个数”为有效排列, 则有效排列的概率为 ()

Arrange 3 identical black balls and 3 identical white balls in a row from left to right. A "valid arrangement" is defined as "from any position, the number of black balls counted to the left is $\geq$ the number of white balls". What is the probability of a valid arrangement? ()

A. $\frac{1}{2}$ B. $\frac{1}{4}$ C. $\frac{1}{5}$ D. $\frac{1}{10}$

13. 已知点 $P_1, P_2, \dots, P_{10}$ 分别是四面体的顶点或棱的中点, 四点组 (P_i, P_j, P_k, P_l) ($1 <$

$i < j < k \leq 10$) 在同一平面上的概率为 ()

Given that points $P_1, P_2, \dots, P_{10}$ are respectively the vertices or midpoints of the edges of a tetrahedron. What is the probability that the four-point group (P_1, P_i, P_j, P_k) ($1 < i < j < k \leq 10$) lies on the same plane? ()

A. $\frac{2}{5}$ B. $\frac{3}{7}$ C. $\frac{4}{9}$ D. $\frac{5}{11}$

14. 东西方向有 4 条道路, 南北方向有 5 条道路, 从点 A 出发沿最短道路行走至 C 点, 中途不经过点 P 的概率是 ()

There are 4 roads in the east-west direction and 5 roads in the north-south direction. Starting from point A , walk along the shortest path to point C . What is the probability of not passing through point P midway? ()

A. $\frac{3}{7}$ B. $\frac{4}{9}$ C. $\frac{5}{11}$ D. $\frac{6}{13}$

15. 将号码 1~9 的九个小球放入袋中, 甲摸出号码 a 放回后, 乙摸出号码 b , 则不等式 $a - 2b + 10 > 0$ 成立的概率为 ()

Put nine small balls numbered 1~9 into a bag. After A draws a ball with number a and puts it back, B draws a ball with number b . What is the probability that the inequality $a - 2b + 10 > 0$ holds? ()

A. $\frac{52}{81}$ B. $\frac{59}{81}$ C. $\frac{60}{81}$ D. $\frac{61}{81}$

16. 一个口袋有 5 个小球 (2 红、2 白、1 黑), 依次摸出 5 个球, 相邻两个小球颜色均不同的概率是 ()

A pocket has 5 small balls (2 red, 2 white, 1 black). Draw 5 balls in sequence. What is the probability that adjacent balls have different colors? ()

A. $\frac{2}{5}$ B. $\frac{1}{5}$ C. $\frac{3}{5}$ D. $\frac{4}{5}$

17. 圆上 27 个点将圆等分, 任取 3 个不同的点, 顺次连接形成正三角形的概率为 ()

30 points on a circle divide the circle equally. Take 3 different points randomly and connect them in sequence. What is the probability of forming an equilateral triangle? ()

A. $\frac{1}{145}$ B. $\frac{1}{95}$ C. $\frac{1}{57}$ D. $\frac{1}{28}$

18. 生物实验室有 5 只兔子, 其中 3 只测量过某项指标, 从 5 只中随机取出 3 只, 恰有 2 只测量过该指标的概率为 ()

A biology laboratory has 5 rabbits, 3 of which have been measured for a certain indicator. Take 3 rabbits randomly from the 5. What is the probability that exactly 2 of them have been measured for the indicator? ()

A. $\frac{2}{3}$ B. $\frac{3}{5}$ C. $\frac{2}{5}$ D. $\frac{1}{5}$

19. 盒中装有 3 个红球、2 个黄球, 随机取出 2 个球, 所取 2 个球颜色不同的概率等于 ()

A box contains 3 red balls and 2 yellow balls. Take 2 balls randomly. What is the probability

that the two balls are of different colors? ()

- A. $\frac{3}{5}$ B. $\frac{2}{5}$ C. $\frac{1}{5}$ D. $\frac{4}{5}$

20. 三位同学参加跳高、跳远、铅球项目比赛，每人选择其中两个项目，则有且仅有两人选择的项目完全相同的概率是 ()

Three students participate in high jump, long jump, and shot put competitions. Each student chooses two of the events. What is the probability that exactly two students choose exactly the same two events? ()

- A. $\frac{1}{3}$ B. $\frac{2}{3}$ C. $\frac{1}{2}$ D. $\frac{3}{4}$

二、非选择题

II. Non-Multiple-Choice Questions

1. 判断下列解答是否正确，并说明理由：

Judge whether the following solution is correct and explain the reason:

某运动员连续两次飞碟射击，用 y 表示命中、 n 表示未命中，样本空间 $\Omega = yy, yn, ny, nn$ ，故“两次都命中”的概率为 0.25。

An athlete shoots at clay pigeons twice in a row. Let y represent a hit and n represent a miss. The sample space is $\Omega = yy, yn, ny, nn$, so the probability of "hitting both times" is 0.25.

2. 从 52 张扑克牌（不含大小王）中随机抽 1 张，计算下列事件的概率：

Draw 1 card randomly from 52 poker cards (excluding jokers). Calculate the probability of the following events:

(1) 抽到的牌是 7; (2) 抽到的牌不是 7; (3) 抽到的牌是方片; (4) 抽到 J 或 Q 或 K; (5) 抽到的牌既是红心又是草花; (6) 抽到的牌比 6 大比 9 小; (7) 抽到的牌是红花色; (8) 抽到的牌是红花色或黑花色。

(1) Drawing a 7; (2) Not drawing a 7; (3) Drawing a diamond; (4) Drawing a J, Q, or K; (5)

Drawing a card that is both a heart and a club; (6) Drawing a card larger than 6 and smaller than 9; (7) Drawing a red-colored card; (8) Drawing a red-colored card or a black-colored card.

3. 从 0~9 这 10 个数中随机选择一个数，求下列事件的概率：

Choose a number randomly from 0~9. Find the probability of the following events:

(1) 这个数平方的个位数字为 1; (2) 这个数的四次方的个位数字为 1。

(1) The units digit of the square of this number is 1; (2) The units digit of the fourth power of this number is 1.

4. 将一枚均匀的硬币投掷 6 次，求正面出现的次数比反面出现的次数多的概率。

Toss a fair coin 6 times. Find the probability that the number of heads is greater than the number of tails.

5. 在 1,2,...,2006 中随机选取三个数，求能构成递增等差数列的概率。

Choose three numbers randomly from 1,2,...,2006. Find the probability that they can form an increasing arithmetic sequence.